

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

Course: S.Y. B. Tech. Branch: Mechanical Engineering Semester: III

Subject Code & Name: BTMES304 Material Science and Metallurgy

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Write Classification of Engineering materials.	CO1	06
B) Explain Imperfection in crystal Structure.	CO1	06
C) What are the types of impact test? Explain the Procedure of Charpy Impact test.	CO1	06
Q.2 Solve Any Two of the following.		
A) What is a TTT diagram? What is its significance and use?	CO2	06
B) Classification of steel on the basis of location on Fe-Fe ₃ C equilibrium diagram (Hypo and Hyper eutectoid steels)	CO2	06
C) Write AISI Specification for steels.	CO2	06
Q. 3 Solve Any Two of the following.		
A) Explain the various steps of the heat treatment and the purpose of each step.	CO3	06
B) Which type of cooling media used in heat treatment list them and explain.	CO3	06
C) Write short note on Quenching and Hardenability	CO3	06
Q.4 Solve Any Two of the following.		
A) What is microscopy? Write down the steps of specimen preparation.	CO4	06
B) Explain the procedure of spark test.	CO4	06
C) Write short note on Optical metallurgical microscope.	CO4	06
Q. 5 Solve Any Two of the following.		
A) What is Refinement of grain size?	CO5	06
B) Write short note on Solid solution strengthening	CO5	06
C) Write the procedure of Dye Penetrant inspection test.	CO5	06

***** End *****